

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110
(An Autonomous Institution, Affiliated to Anna University, Chennai)

Date: 22.06.2026

Course Code	UEC3942	Course Name	Introduction to Sensors and Actuators				
Course Type	Theory	Course Category	Open Elective (OE)	L	T	P	C
				3	0	0	3
Regulation	2021		Academic Year	2026-2027			
Degree and Branch			B.E. (CSE, EEE, IT & MECH)				
Semester			VI				
Department Offering the Course			Electronics and Communication Engineering				

Preamble:

This subject deals with the study and understanding of different types of sensors and actuators. It defines the characteristics of micro sensors and micro actuators and gives an in-depth knowledge about the operation of various sensors and its applications in the real time scenarios. The subject also deals with the characterization of MEMS/NEMS devices and its available packaging techniques.

Course Objectives:

- To study different types of sensors and actuators
- To gain knowledge on type of optical sensors to be used for practical applications
- To understand the design concepts of micro sensors and micro actuators
- To understand the packaging and characterization of MEMS/NEMS

Syllabus

UNIT I STRAIN, PRESSURE AND TEMPERATURE 8

Introduction, Stress & Pressure sensors: Resistive strain gauge, piezoelectric strain gauge, characteristics. Fiber-optic sensor, Pressure gauges. Temperature Sensors: Bimetallic strip, thermocouples, Resistance thermometers, thermistors, Pyroelectric detector.

UNIT II OPTICAL SENSORS 9

Color temperature, light flux, photo sensors, photomultiplier, photo resistor and photoconductors, photodiodes, phototransistors, photovoltaic devices, fiber-optic applications, light transducer, solid-state transducers and liquid crystal devices.

UNIT III MICRO SENSORS 9

Acoustic sensor – Quartz crystal microbalance, Surface acoustic wave, Flexural plate wave, shear horizontal; Vibratory gyroscope, Capacitive and Piezoresistive Pressure sensors.

UNIT IV MICRO ACTUATORS 9

Electrostatic actuators – parallel plate capacitor, Interdigitated finger capacitors; piezoelectric actuators, Thermal actuators, Actuators using shape memory alloys, Microgrippers, Micromotors, Microvalves, Micropumps.

UNIT V PACKAGING AND CHARACTERIZATION OF MEMS AND NEMS 9

Micro / nano systems packaging, Essential packaging technologies, Selection of packaging materials; Nano material characterization - SEM, TEM, AFM, STM, XRD.

Course Outcomes:

- CO1 Select sensors for the measurement of physical variables for real time applications
- CO2 Use sensors for optoelectronic applications
- CO3 Elucidate and apply sensors for engineering applications
- CO4 Elucidate and apply actuators for engineering applications
- CO5 Examine the packaging and characterization requirements

Textbooks:

1. Ian R Sinclair, Sensors and Transducers, Newnes publishers, Third Edition, 2001 (Unit I & II).
2. Tai Ran Hsu, MEMS and Microsystems Design and Manufacture, Tata McGraw Hill, First Edition, 2002 (Unit III to V).

References:

1. Doebelin E O, Measurement Systems, Application and Design, McGraw Hill, Fifth Edition, 2004.
2. Murty B.S, Shankar P, Raj B, Rath, B.B, Murday J, Textbook of Nanoscience and Nanotechnology, Springer publishing, 2013.
3. Vinod Kumar Khanna, Nanosensors: Physical, Chemical, and Biological, CRC press, First Edition, 2012.
4. Chang Liu, Foundations of MEMS, Pearson education India limited, Second Edition, 2006.
5. Ramon Pallas Areny and John G. Webster, Sensors and Signal conditioning, John Wiley and Sons, Second Edition, 2000.

Lesson Plan

Sl. No.	Course Content	Course Outcome	Knowledge level	Proposed Period & Number of Hours to be handled	Actual Period & Number of Hours handled
Unit I STRAIN, PRESSURE AND TEMPERATURE					
1	Introduction, Stress & Pressure sensors	CO1	K2	2	
2	Resistive strain gauge, piezoelectric strain gauge, characteristics	CO1	K2	2	
3	Fiber-optic sensor, Pressure gauges	CO1	K2	2	
4	Temperature Sensors: Bimetallic strip, thermocouples, Resistance thermometers, thermistors, Pyroelectric detector	CO1	K2	3	
Unit II OPTICAL SENSORS					
1	Color temperature, light flux	CO2	K2	2	
2	Photo sensors, photomultiplier, photo resistor and photoconductors	CO2	K3	2	
3	Photodiodes, phototransistors, photovoltaic devices	CO2	K3	3	
4	Fiber-optic applications, light transducer, solid-state transducers and liquid crystal devices	CO2	K3	2	
Unit III MICRO SENSORS					
1	Acoustic sensor – Quartz crystal microbalance, Surface acoustic wave	CO3	K3	2	
2	Flexural plate wave, shear horizontal	CO3	K3	2	
3	Vibratory gyroscope	CO3	K3	3	
4	Capacitive and Piezoresistive Pressure sensors	CO3	K3	2	
Unit IV MICRO ACTUATORS					
1	Electrostatic actuators – parallel plate capacitor, Interdigitated finger capacitors	CO3	K2	2	
2	Piezoelectric actuators, Thermal actuators, Actuators using shape memory alloys	CO3	K2	3	
3	Microgrippers, Micromotors	CO3	K3	2	
4	Microvalves, Micropumps	CO3	K3	2	

Unit V PACKAGING AND CHARACTERIZATION OF MEMS AND NEMS					
1	Micro / nano systems packaging	CO4	K3	3	
2	Essential packaging technologies	CO4	K3	2	
3	Selection of packaging materials	CO4	K3	3	
4	Nano material characterization - SEM, TEM, AFM, STM, XRD	CO4	K3	2	
Content beyond Syllabus (Theory) - Topics					
1	Quantum-based optical sensors	CO4	K4	1	
2	Smart Sensors and Actuators	CO4	K4	1	

Mapping of Course Outcomes with Program Outcomes

Course Outcomes	Program Outcomes and Performance Indicators																							
	1	PI	2	PI	3	PI	4	PI	5	PI	6	PI	7	PI	8	PI	9	PI	10	PI	11	PI	12	PI
CO1: Select sensors for the measurement of physical variables for real time applications.	2	1.4.1	2	2.2.1	2	3.2.2	1	4.3.1	1	5.1.2	2	6.1.1	1	7.2.2			1	9.1.2	1	10.1.2			1	12.2.2
CO2: Use sensors for optoelectronic applications.	2	1.4.1	3	2.2.1	3	3.2.2	3	4.3.1	3	5.1.2	2	6.1.1	2	7.2.2			1	9.1.2	1	10.1.2			1	12.2.2
CO3: Apply actuators for engineering applications.	2	1.4.1	3	2.2.1	3	3.2.2	3	4.3.1	3	5.1.2	2	6.1.1	2	7.2.2			1	9.1.2	1	10.1.2			1	12.2.2
CO4: Explore the packaging and characterization requirements.	1	1.4.1	2	2.2.1	3	3.2.2	1	4.3.1	2	5.1.2			1	7.2.2					1	10.1.2			1	12.2.2

Mapping of Course Outcomes with Program Specific Outcomes			
Course Outcomes	Program Specific Outcomes		
	PSO 1	PSO 2	PSO 3
CO1: Select sensors for the measurement of physical variables for real time applications.	2		
CO2: Use sensors for optoelectronic applications.	2	2	3
CO3: Apply actuators for engineering applications.	2	2	3
CO4: Explore the packaging and characterization requirements.			3

Program Outcomes:		
PO1	Engineering knowledge: Acquire the knowledge of mathematics, science, engineering fundamentals, and electronics and communication engineering, with an ability to understand, analyze and apply to the solution of engineering problems.	
PO2	Problem analysis: Identify, formulate, research literature, analyse and solve electronics and communication engineering problems.	
PO3	Design/development of solutions: Design solutions for electronics and communication engineering problems with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	
PO4	Conduct investigations of complex problems: Design and conduct experiments, analyse and interpret data and synthesize information to provide valid conclusions.	
PO5	Modern tool usage: Apply appropriate techniques, resources, and modern hardware and software engineering tools to solve electronics and communication engineering problems.	
PO6	The engineer and society: Apply reasoning to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	
PO7	Environment and sustainability: Demonstrate the understanding of the impact of the professional engineering solutions in societal and environmental contexts, and need for sustainable development.	
PO8	Ethics: Demonstrate the knowledge of professional and ethical responsibilities.	
PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	
PO10	Communication: Comprehend and communicate confidently and effectively in both verbal and written form.	
PO11	Project Management and Finance: Apply the engineering and management principles for efficient project management.	
PO12	Life-long Learning: Recognize the need and acquire confidence for independent and life-long learning.	
Program Specific Outcomes:		
PSO1	Design, develop and analyze electronic systems through application of relevant mathematics and engineering principles	
PSO2	Design, develop and analyze communication systems through application of fundamentals from wireless communication network, signal processing, and RF & electromagnetics	
PSO3	Adapt to emerging electronics and communication technologies and develop innovative solutions for existing and newer problems	
3: Strong	2: Significant	1: Reasonable

Justification for Course Outcome – Program Outcome Mapping:

The Course Outcomes CO1, CO2 and CO3 have a strong to significant mapping with PO1, PO2 and PO3, as the course provides comprehensive engineering knowledge of sensors and actuators, their operating principles, characteristics, and real-time applications. Students analyse different sensing mechanisms (mechanical, thermal, optical and micro-sensors) and actuator types, enabling them to identify, formulate and solve engineering problems related to measurement, control and automation systems. The design-oriented content in micro sensors and micro actuators, along with application-based learning, significantly contributes to the development of problem analysis and solution design skills, thereby justifying strong and significant mappings with PO1 to PO3. Additionally, laboratory-oriented discussions and case-based examples support significant mapping with PO4 and PO5 by encouraging experimental understanding, data interpretation and the use of modern engineering tools.

Course Outcome CO4 have a reasonable and significant mapping with PO4 and PO5 due to the detailed coverage of MEMS/NEMS packaging technologies and advanced characterization techniques such as SEM, TEM, AFM and XRD. This outcome enhances students' ability to investigate complex engineering problems and apply modern tools for system analysis and validation. The course also demonstrates reasonable to significant mapping with PO6, PO7, and PO12 by creating awareness of safety, sustainability, and the need for continuous learning in sensor-based engineering solutions. Team-based discussions, technical reporting and presentation components contribute to reasonable mapping with PO9 and PO10, while exposure to real-world applications provides foundational understanding related to project planning and lifelong learning, thereby supporting the overall CO-PO alignment.

Assessment of Course Outcomes:

Assessment Tool	Unit	CO1	CO2	CO3	CO4
Assessment - I	I & II	√	√		
Assessment - II	III & IV			√	
Assignment - I	I & II	√	√		
Assignment - II	V				√

(put √ mark in the appropriate box)

Remedial measures for Gap in Course Outcomes: No Gap is observed in the course outcomes

Continuous Assessment Test				End Semester Examination
Assessment - I (100 Marks)		Assessment - II (100 Marks)		Total External Assessment (100 Marks) shall be converted to 60 Marks
Theory		Theory		
Individual Assignment / Case Study / Seminar / Mini Project	Written Test	Individual Assignment / Case Study / Seminar / Mini Project	Written Test	
40 Marks	60 Marks	40 Marks	60 Marks	
Total Internal Assessment (200 Marks) shall be converted to 40 Marks				

- ✓ A minimum of two internal assessments will be conducted as a part of continuous assessment.
- ✓ Each internal assessment is to be conducted for 100 marks and will have to be distributed in two parts viz., Individual Assignment/Case study/Seminar/Mini project and Test with each having a weightage of 40% and 60% respectively.
- ✓ The tests shall be in written mode.
- ✓ The total internal assessment marks of 200 shall be proportionately converted into a maximum of 40 marks and rounded to the nearest integer.

Dr. C. Annadurai
(Course In-charge)

ISO / IQAC Coordinator

Dr. P. Vijayalakshmi
(HoD / ECE)